EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 3 - SOLUTIONS

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

Chapter 8: 5

$$V_1 = V_2 = V_s = IR_T = 0.6\text{ A}[6\ \Omega \parallel 24\ \Omega \parallel 24\ \Omega] = 0.6\text{ A}[6\ \Omega \parallel 12\ \Omega] = 2.4\text{ V}$$

$$I_2 = \frac{V_2}{R_2} = \frac{2.4\text{ V}}{24\ \Omega} = \mathbf{0.1\ A}$$

$$V_3 = \frac{R_3 V_s}{R_3 + R_4} = \frac{16\ \Omega(2.4\text{ V})}{24\ \Omega} = \mathbf{1.6\ V}$$

6,

a. $I_1 = \dfrac{E}{R_1} = \dfrac{24\text{ V}}{2\ \Omega} = \mathbf{12\ A}$, $I_{R_2} = \dfrac{E}{R_2 + R_3} = \dfrac{24\text{ V}}{6\ \Omega + 2\ \Omega} = \dfrac{24\ \Omega}{8\ \Omega} = 3\text{ A}$

KCL: $I + I_s - I_1 - I_{R_2} = 0$

$I_s = I_1 + I_{R_2} - I = 12\text{ A} + 3\text{ A} - 4\text{ A} = \mathbf{11\ A}$

b. $V_s = E = 24\text{ V}$

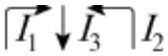
VDR: $V_3 = \dfrac{R_3 E}{R_2 + R_3} = \dfrac{2\ \Omega(24\text{ V})}{6\ \Omega + 2\ \Omega} = \dfrac{48\text{ V}}{8\ \Omega} = \mathbf{6\ V}$

13. a. Conversions: $I_1 = E_1/R_1 = 9\text{ V}/3\ \Omega = \mathbf{3\ A}$, $R_1 = \mathbf{3\ \Omega}$
$I_2 = E_2/R_2 = 20\text{ V}/2\ \Omega = \mathbf{10\ A}$, $R_2 = \mathbf{2\ \Omega}$

b. $I_T\downarrow = 10\text{ A} - 3\text{A} = 7\text{ A}$, $R_T = 3\ \Omega \parallel 6\ \Omega \parallel 2\ \Omega$
$= 2\ \Omega \parallel 2\ \Omega$
$= 1\ \Omega$

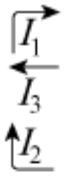
$V_{ab} = -I_T R_T = -(7\text{ A})(1\ \Omega) = \mathbf{-7\ V}$
+−

c. $I_3\uparrow = \dfrac{7\text{ V}}{6\ \Omega} = \mathbf{1.17\ A}$

17. 

$10 - I_1\ 5.6\text{ k}\Omega - I_3\ 2.2\text{ k}\Omega + 20 = 0$
$-20 + I_3\ 2.2\text{ k}\Omega + I_2\ 3.3\text{ k}\Omega - 30 = 0$
$I_1 + I_2 = I_3$

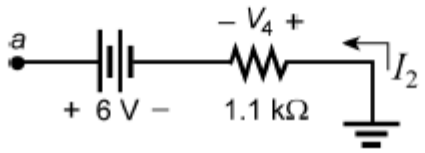
$I_1 = I_{R_1} = \mathbf{1.45\ mA}$, $I_2 = I_{R_2} = \mathbf{8.51\ mA}$, $I_3 = I_{R_3} = \mathbf{9.96\ mA}$

18. a. 

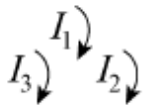
$-1.2\text{ k}\Omega\ I_1 + 9 - 8.2\text{ k}\Omega\ I_3 = 0$
$-10.2\text{ k}\Omega\ I_2 + 8.2\text{ k}\Omega\ I_3 + 6 = 0$
$I_2 + I_3 = I_1$

$I_1 = \mathbf{2.03\ mA}$, $I_2 = \mathbf{1.23\ mA}$, $I_3 = \mathbf{0.8\ mA}$

$I_{R_1} = I_1 = 2.03\text{ mA}$
$I_{R_2} = I_3 = 0.8\text{ mA}$
$I_{R_3} = I_{R_4} = I_2 = \mathbf{1.23\ mA} = I_{9.1\text{k}\Omega}$

b. 

26. a.



$$16 - 4I_1 - 3(I_1 - I_2) - 12 - 4(I_1 - I_3) = 0$$
$$12 - 3(I_2 - I_1) - 10\,I_2 - 15 - 4(I_2 - I_3) = 0$$
$$-16 - 4(I_3 - I_1) - 4(I_3 - I_2) - 7I_3 = 0$$

b. $I_1 = \mathbf{-0.24\ A}$, $I_2 = \mathbf{-0.52\ A}$, $I_3 = \mathbf{-1.28\ A}$

c. $I_{R_5} = I_1 = \mathbf{-0.24\ A}$

39. a.

$$(1\ \Omega + 2\ \Omega + 4\ \Omega)I_1 - 2\ \Omega I_2 - 4\ \Omega I_3 = 12\ \text{V}$$
$$(2\ \Omega + 2\ \Omega + 10\ \Omega)I_2 - 2\ \Omega I_1 - 10\ \Omega I_3 = -20\ \text{V}$$
$$(4\ \Omega + 10\ \Omega + 8\ \Omega)I_3 - 10\ \Omega I_2 - 4\ \Omega I_1 = 20\ \text{V}$$

$$7I_1 - 2I_2 - 4I_3 = 12$$
$$2I_1 - 14I_2 + 10I_3 = 20$$
$$4I_1 + 10I_2 - 22I_3 = -20$$

b. $I_1 = \mathbf{2.38\ A}$, $I_2 = \mathbf{-0.195\ A}$, $I_3 = \mathbf{1.25\ A}$

c. $V_a = (I_1 - I_3)4\ \Omega = (2.38\ \text{A} - 1.25\ \text{A})4\ \Omega = \mathbf{4.5\ V}$
$V_b = I_3 8\ \Omega = (1.25\ \text{A})(8\ \Omega) = \mathbf{10\ V}$

d. $V_{ab} = V_a - V_b = 4.5\ \text{V} - 10\ \text{V} = \mathbf{-5.5\ V}$

41. a.

At V_1: $\Sigma I_i = \Sigma I_o$

$$0 = \frac{V_1}{8\ \Omega} + 12\,\text{A} + I_{6\Omega} \text{ and } V_1 - I6\Omega - 54\ \text{V} - V_2 = 0$$

$$\text{or } I = \frac{V_1 - V_2 - 54\ \text{V}}{6\ \Omega} = \frac{V_1}{6\ \Omega} + \frac{V_2}{6\ \Omega} - 9\ \text{A}$$

$$\text{so that } 0 = \frac{V_1}{8\ \Omega} + 12\ \text{A} + \frac{V_1}{6\ \Omega} - \frac{V_2}{6\ \Omega} - 9\,\text{A}$$

$$\text{or } V_1\left[\frac{1}{8\ \Omega} + \frac{1}{6\ \Omega}\right] - V_2\left[\frac{1}{6\ \Omega}\right] = -12\ \text{A} + 9\ \text{A} = -3\ \text{A}$$

At V_2: $\sum I_i = \sum I_o$

$$I = \frac{V_2}{20\ \Omega} + \frac{V_2}{5\ \Omega}$$

or $$\frac{V_1}{6\ \Omega} - \frac{V_2}{6\ \Omega} - 9\ \text{A} = \frac{V_2}{20\ \Omega} + \frac{V_2}{5\ \Omega}$$

and $$V_2\left[\frac{1}{6\ \Omega} + \frac{1}{20\ \Omega} + \frac{1}{5\ \Omega}\right] - V_1\left[\frac{1}{6\ \Omega}\right] = -9\ \text{A}$$

resulting in $$V_1\left[\frac{1}{8\ \Omega} + \frac{1}{6\ \Omega}\right] - V_2\left[\frac{1}{6\ \Omega}\right] = -3\ \text{A}$$

$$-V_1\left[\frac{1}{6\ \Omega}\right] + V_2\left[\frac{1}{6\ \Omega} + \frac{1}{20\ \Omega} + \frac{1}{5\ \Omega}\right] = -9\ \text{A}$$

---

b. $V_1 = \mathbf{-29.29\ V}$, $V_2 = \mathbf{-33.34\ V}$

c. $I_{20\Omega} \uparrow = \dfrac{V_2}{20\ \Omega} = \dfrac{33.34\ \text{V}}{20\ \Omega} = \mathbf{1.67\ A}$

45. a.

Source Conversion: $I_2 = \dfrac{15\ \text{V}}{3\ \Omega} = 5\ \text{A}$, $R_p = R_1 = 3\ \Omega$

At V_1: $\sum I_i = \sum I_o$

$$5\ \text{A} = \frac{V_1}{3\ \Omega} + \frac{V_1 - V_2}{6\ \Omega} + \frac{V_1 - V_3}{6\ \Omega}$$

At V_2: $\sum I_i = \sum I_o$

$$\frac{V_1 - V_2}{6\ \Omega} = \frac{V_2}{4\ \Omega} + 3\ \text{A} + \frac{V_2 - V_3}{5\ \Omega}$$

At V_3: $\sum I_i = \sum I_o$

$$\frac{V_1 - V_3}{6\ \Omega} + \frac{V_2 - V_3}{5\ \Omega} = \frac{V_3}{7\ \Omega}$$

Rewritten:

$$V_1\left[\frac{1}{3\ \Omega}+\frac{1}{6\ \Omega}+\frac{1}{6\ \Omega}\right]-\frac{1}{6\ \Omega}V_2-\frac{1}{6\ \Omega}V_3=5\text{ A}$$

$$V_2\left[\frac{1}{6\ \Omega}+\frac{1}{4\ \Omega}+\frac{1}{5\ \Omega}\right]-\frac{1}{6\ \Omega}V_1-\frac{1}{5\ \Omega}V_3=-3\text{ A}$$

$$V_3\left[\frac{1}{6\ \Omega}+\frac{1}{5\ \Omega}+\frac{1}{7\ \Omega}\right]-\frac{1}{5\ \Omega}V_2-\frac{1}{6\ \Omega}V_1=0$$

---

b. V_1 = **7.24 V**, V_2 = **–2.45 V**, V_3 = **1.41 V**

c. $V_{5\Omega}$ = $V_3 - V_2$ = 1.41 V – (–2.45 V) = **3.86 V**

48. a. $\overset{-}{\circ}\!\!\overset{+}{V_1}$ $\circ V_2$

$\circ V_3$

At V_1: $\sum I_i = \sum I_o$

$$0 = 5\text{ A} + \frac{V_1}{2\ \Omega} + \frac{V_1 - V_3}{6\ \Omega}$$

At V_2: $\sum I_i = \sum I_o$

$$5\text{ A} = 2\text{ A} + \frac{V_2}{4\ \Omega}$$

At V_3: $\sum I_i = \sum I_o$

$$\frac{V_1 - V_3}{6\ \Omega} + 2\text{A} = \frac{V_3}{5\ \Omega}$$

Rewritten:

$$V_1\left[\frac{1}{2\ \Omega} + \frac{1}{6\ \Omega}\right] - \frac{1}{6\ \Omega}V_3 = -5\ \text{A}$$

$$V_2\left[\frac{1}{4\ \Omega}\right] = 5\ \text{A} - 2\ \text{A}$$

$$V_3\left[\frac{1}{6\ \Omega} + \frac{1}{5\ \Omega}\right] - \frac{1}{6\ \Omega}V_1 = 2\text{A}$$

---

b. $V_1 = \mathbf{-6.92\ V}$, $V_2 = \mathbf{12\ V}$, $V_3 = \mathbf{2.3\ V}$

c. $I_{4\Omega} = \dfrac{V_2}{4\ \Omega} = \dfrac{12\ \text{V}}{4\ \Omega} = \mathbf{3\ A}$